Switching Module-XC (SM-XC) Growth with CM3

Excerpt from an installation procedure. Selected sections, figures, and one table are shown; step, figure, and table numbers are kept from the full procedure. Company names, product brand, document numbers, internal links, and a sample office name have been removed.

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1. General Information

1.1 Purpose

The purpose of this procedure is to grow a Switching Module-XC (SM-XC), Switching Module Unit Model 6 (SMU6) into an operating switch office without interfering in customer calls or causing loss of service. This section is applicable to those offices using the Communication Module Model 3 (CM3).

NOTE: Message Handler (MH PIPE) Assignments [RCV View 17.20] are automatically generated for the identified growth SM-XC SMU6 when this procedure is performed

SCOPE

This section supports growth of one SM-XC Switching Module Unit Model 6 (SMU6) into an operating switch office. This section is applicable to those offices using the Communication Module Model 3 (CM3).

1.4 Test Equipment, Tools, and Supplies

The following test equipment, tools, and supplies (or the equivalents) are required to perform this section. Vendor test equipment, tools, and supplies may not, in all cases, be available to customers performing their own installation. In all cases, it is suggested that the customer utilize equivalent tools or test sets as a primary source.

ITE-7150	MT-RJ Loopback Plug And Coupler (for use with BKXL1 paddle boards)
ITE-7093	SC-SC Loopback Plug and Coupler (for use with BKXL2 paddle boards)

1.5 Safe Stop Points

The steps in the procedure labeled “Safe Stop Point” are points where progress in the execution of the procedure may be suspended temporarily without causing degradation in the operation of the switch. Execution of the procedure can be halted at one of these points only if all prior steps have been successfully completed.

1.6 Terminology

The term SM-XC refers to the category of switching module, like SM or SM-2000. The term SMU6, refers to the switching module processor unit type associated with the SM-XC, like SMPU4 or SMPU5 is to SM-2000.

The term SM-XC will be used in this procedure to specify general information. The term SMU6 will be used whenever work is being done on the hardware.

1.7 Hardware Description

The Switching Module-XC (SM-XC) Switching Module Unit Model 6 (SMU6) is a duplex processor unit that contains both the module controller (MC) and time slot interchanger (TSI) circuit packs. The SMU6 contains two service groups, 0 and 1. The SMU6 consists of two backplanes, one housing the module controller (MC) and digital service circuits (DSCs), and the other housing the time slot interchange (TSI) complex. The MC and DSC backplane uses Fastech technology, and the TSI backplane uses Metral technology.

The SMU6 is located in an SM-XC cabinet. Up to three SMU6 units are allowed within the cabinet. The SMU6s are equipped in the cabinet from bottom to top. Refer to Figure 1 for a view of the SM-XC cabinet housing 3 SMU6s. The first SMU6 equipped is the only processor in the cabinet that has fan alarms tied to it. Refer to Figure 2 for a front view of the SMU6, and Figure 3 for a rear view of the SMU6.

The SM-XC SMU6 service groups behave the same way as MCTSI service groups on SM-2000s. For example, if service group 0 of the SMP is removed from service then group 0 of the TSI is removed from service. The same logic applies to power, for example powering down MC service group 1 will power down

TSI service group 1.

The MC contains one application controller (APC), up to two message handlers (MHPPCs), one optional application controller for OXU (APCO), one microprocessor core pack (CORE700 or CORE970), one bus service node (BSN) pack, one power and control/display (PWR/CD) pack, one control interface (CI2), and one packet interface (PI2). The DSC contains circuit packs that provide local digital services function (LDSF - UN363: DSC3/LDSF2- UN590: DSC4), global digital services function (GDSF - UN363: DSC3/GDSF2-UN590: DSC4) or service announcement system (SAS - UN750) function. DSC growth is covered in the related peripheral unit procedure.

The TSI contains three types of circuit packs, the TSISCOM, TSIS3X, and one extended data expansion (XDX) pack.

The TSISCOM pack distributes control messages from the SMP and clock pulses to the time slot interchanger slice (TSIS) and XDX circuits. The TSISCOM pack also provides the interface for the primary network link interfaces (NLIs) and the BKCT1 paddleboard.

The BKCT1 paddleboard houses the control time slot number select (CTSNS) DIP switches, external timing selection DIP switches, external timing interfaces, MCP and XDX links, STCL, power interlock, and -48 volt power. Refer to Figure 4 for a view of the BKCT1 paddleboard.

There are two CTSNS DIP switches on each BKCT1 paddleboard that are used to store the TSI link byte numbers that will be used to transport the CTSs from CM side 1 and CM side 0. One DIP switch holds the CTS number for odd links (labeled - Odd Control Time Slot Number) and the other holds the CTS number for even links (labeled - Even Control Time Slot Number). The value stored in corresponding DIP switches must be the same (i.e. the value of CTSNS 1 on service group 0 must be the same value as CTSNS 1 on service group 1). Refer to Table C for setting of DIP switches for CTS numbers.

The TSIS3X circuit pack has the circuitry for three TSIS partitions. Each TSIS partition has four ports for NLIs and PLIs.

The NLI paddleboards terminate at the TSISCOM circuit pack or at the TSIS3X circuit pack. Two types of paddleboards may be equipped. The BKXL1 paddleboard has a MT-RJ connector for the NCT2 links, and is only valid on the CM3. The BKXL2 paddleboard has an ST connector for NCT2 links and is valid on a CM2 or CM3.

The BKXL1 and BKXL2 paddleboards can be used for either PLIs or NLIs. Two small DIP switches on the paddleboard select the mode. For NLI mode, both DIP switches are set in the down position. For PLI mode, both DIP switches are set in the up position. Refer to Figure 5 for a view of a BKXL1 and BKXL2 paddleboard set in NLI mode.

1.9 Prerequisite Conditions

1. The required CM3 hardware which supports the identified TMS Links must be equipped and in the Operational state. If the hardware is equipped and associated TMS Links are available they will be in the Grow equipage state. Verify identified TMS Links are available in the Grow equipage state. See Step 2.4.1.

NOTE: Refer to the growth installation sequencing procedure for growth installation sequencing information to grow this equipment if required.

2. Assure that all CNs and Software Updates relating to SM-XC equipment have been applied and are up to date.

WARNING: Verify Software Update levels: ALL Software Updates must be official or backed out. No Software Updates will be in the soak or applied state coincidental with this procedure.

3. The office must be on the minimum required software release and software update for wireline or wireless offices. The CORE970 requires a later release and update.
4. If using the MGRCV tool to apply recent change data:

- All elements of the MGRCV Package must be downloaded and installed on the switch as described in the related handbook procedure.
- Paper copies of each associated completed GRCV form (or access to printable files containing each GRCV form) is required. This Recent Change data is needed, especially after data insert operations, so that a visual comparison can be performed between the data in the switch and that on the GRCV form.

5. Verify that XLI paddleboards (BKXL1 or BKXL2) are set to NLI Mode (Both Dip switches are set to the “down” position). See Figure 5.

NOTE: If any of the BKXL (1 or 2) packs have tape over the NLI/PLI switch settings, remove it before setting the switches.

6. Verify that the OXU APCO strap is not equipped. This strap is located at 121-005 on both MCTSI 0 and MCTSI 1. Remove this strap if it is equipped. If an OXU is grown on after the SM Core growth, be sure to install during that procedure.
7. For the SM being grown, obtain the desired REX schedule from the customer. Make sure all of the necessary information from RC/V view 8.3 is available. The SM REX schedule will be inserted before the SM is made operational. If this is not done, a craft assert will occur every day at midnight.

NOTE: If a schedule is not available or the customer does not want to insert a REX schedule for the SM at this time, default values for RC/V view 8.3 are listed during this procedure.

Refer to the "REX Description" section of the Routine Operations and Maintenance Procedures Manual. To obtain a suggested REX schedule for the office, use the automated REX schedule tool. Instructions for this tool are in that manual.

1.10 Podcast

Handbook updates and training information for this job are published through the Installation Training and Documentation podcast.

1.11 Training

A web-based SM-XC installation course is available for installers performing this job.

1.12 Job Aids

Job aids that support this job are available on the job aids web page.

2. Switching Module-XC Growth Procedure

2.1 Time Slot Allocation

2.1.1 What Generic is the switch on?

- If release 21.1 or later, continue with the next step.
- If NOT, go to Step 2.2.

2.1.2 Verify the number of time slots allocated. At the MCC, type and enter:

```
OP:QLPSTS;
```

```
Response:
```

+++ OFFICE1 06-04-07 11:04:30 MTCE 6444 #000168 >

13072M ORIGINATING COMMAND # = 000007.0016

13073 OP QLPTS

13074	NETWORK	QTMSLNK	AVAIL	USED	TOTAL
13075	0	0	206	50	256
13076	0	1	216	40	256
13077	0	2	178	78	256
13078	0	3	210	46	256
13079	1	0	206	50	256
13080	1	1	216	40	256
13081	1	2	178	78	256
13082	1	3	210	46	256

13083 END OF REPORT #000168 ++-

2.1.3 There must be at least 32 QLPS time slots available in one of the four QTMSLNKs. If the number in the AVAIL column is not 32 or greater in one of the four QTMSLNKs, Seek Technical Assistance.

2.2 Identify the Growth SM-XC

2.2.1 Determine the Growth SM-XC number.

2.2.2 Determine minimum hardware equipage requirement is present for identified growth SM-XC. Refer to the growth installation sequencing procedure for growth installation sequencing information.

2.2.3 Determine the vertical position of MCTSI 0 and MCTSI 1. Up to 3 SMU6s are allowed in an SM-XC cabinet.

1. Determine the Vertical Position (VERPOS) of the SM-XC from GRCV form 18.1.
2. Determine the cabinet EQL of MCTSI side 0 (V0) and MCTSI Side 1 (V1). Use the table below to correlate the VERPOS field value with the cabinet EQL. Values V0 and V1 will be used throughout this procedure.

18.1 VERPOS	MCTSI 0 Cabinet EQL (V0)	MCTSI 1 Cabinet EQL (V1)
6	10	19
32	36	45
49	53	62

NOTE: The VERPOS field value is the vertical equipment location of the bottom of the SMU6. From this value, an offset is added to get the software (TLP) location of the unit. This value can be slightly different than the cabinet EQL.

3. If the 18.1 VERPOS field value does not appear in the table above, use these guidelines:

— Verify that the VERPOS field value is correct. The value should reference the bottom of the SMU6.

— For MCTSI 0 Cabinet EQL (V0), add 4 to the VERPOS field value.

— For MCTSI 1 Cabinet EQL (V1), add 13 to the VERPOS field value.

NOTE: Valid Cabinet EQLs are: 10, 19, 28, 36, 45, 53, and 62. Refer to Figure 1 for a view of the SM-XC cabinet.

4. Record the SMU6 MCTSI 0 (V0) and MCTSI 1 (V1) Cabinet EQLs in the table below.

Cabinet EQL	Value
V0	
V1	

2.4 Establish Initial Conditions

2.4.1 Verify the Communication Module TMS Links are equipped and in the Grow equipage state.

CAUTION: If the required eight TMS Links are not found in the Grow equipage state, exit this SM-XC growth procedure and perform the required TMS Shelf/Link growth first. If you would like to generate an output report for all TMS Links which are in the Grow equipage state, at MCC type and enter: OP:EQSTAT,GROWTH;

1. At MCC, type and enter:

```
OP:EQSTAT,TMSLNK=a-0-c;

Where:
    a = ONTC side number.

    0 = TMSFP number.
    c = TMS Link number, the identified even or odd links [They are part of the identified GRCV(s) 1

Response: OP EQSTAT CM output report is generated for the TMS Link(s) defined on the
     17.12 GRCV form.
```

2. Repeat the above command as required and examine the system outputs to verify that each of the eight Time Multiplex Switch (TMS) Links required for the growth SM-XC are defined in the system with the Grow equipage state. Determined by matching the eight growth GRCV 17.12 forms to system outputs.
3. Sequence GRCV 17.12 forms by their KEY attributes; and their association with the GRCV 18.16 form. Record/write the LINK number attributes in sequence for the eight 17.12 forms below: Also, record/write these values within this procedure at Step 2.29.1, Step 2.37.1, Step 2.61.1, and Step 2.64.1 as required.

GRCVs 17.12 RCV 18.16 - EVEN NLI 1. SIDE 0, TMSFP NUM 0, LINK ___<--Even Link 0 number is the---------- TMS LINK 0 att. value 2. SIDE 0, TMSFP NUM 0, LINK___<--add +1 to above for Odd Link 0 number 3. SIDE 0, TMSFP NUM 0, LINK___<--Even Link 1 number is the---------- TMS LINK 1 att. value 4. SIDE 0, TMSFP NUM 0, LINK___<--add +2 to Odd Link number above for Odd Link 1 number - ODD NLI 5. SIDE 1, TMSFP NUM 0, LINK___<--Even Link 0 number is the---------- TMS LINK 0 att. value 6. SIDE 1, TMSFP NUM 0, LINK___ (numbered same as above NLIs) 7. SIDE 1, TMSFP NUM 0, LINK___ TMS LINK 1 att. value 8. SIDE 1, TMSFP NUM 0, LINK___

2.4.2 At the MCC observe the Summary Status Area for a SYS NORM indication.

- If the SYS NORM indicator is highlighted in green, go to Step 2.4.3.
- If the SYS NORM indicator is not highlighted in green, continue.

1. If a SYS NORM indication is not observed, determine the reason why with the appropriate input command. At the MCC, type and enter:

   ```
   OP:SYSSTAT;
   ```

   ```
   Response: OP SYSSTAT SUMMARY report printed on the ROP.
   ```

2. Evaluate the reports and any SM Status pages (other than growth SM-XC) to determine if system status is acceptable to continue. Specific attention is required for areas that can disrupt call processing. If any problems exist that could impact the execution of the procedure or call processing they must be resolved prior to continuing.

2.4.3 Inhibit routine exercises for the AM. At the MCC, type and enter:

CAUTION: If DMQ:SRC=REX is inhibited, it must be allowed at the conclusion of the growth procedure. It must also be allowed if pausing at a procedure Safe Stop Point for an extended period of time.

```
INH:DMQ:SRC=REX;
```

```
Response: OK
```

2.4.4 Inhibit routine exercises for the CM. At MCC, type and enter:

```
INH:REX:CM;
```

```
Response: OK
```

2.4.5 Inhibit routine exercises for the equipped SMs. At MCC, type and enter:

```
INH:REX,SM=a&&b;
```

```
Where: a = Lowest numbered SM.
     b = Highest numbered SM.
```

```
Response: OK
```

2.4.6 Inhibit automatic reorganization for the switch. At MCC, type and enter:

NOTE: If a Safe Stop Point is reached, ALW-REORG may be resumed. Enter the following to allow reorganization: ALW:REORG; Be certain to inhibit the reorganization before continuing the procedure.

```
INH:REORG;
```

```
Response: OK
```

2.4.7 Record the ODD backup schedule if one exists. At MCC, type and enter:

```
OP:BKUPSTAT;
```

```
Response:
    OP BKUST
[FULL][AM][NRODD=a[TO b]][RODD=a]EVERY=c AT=ffff
```

```
Where:
    a = Lowest numbered SM.

    b = Highest numbered SM.
    c = Number of days between each automatic execution of the ODD backup.
    ffff = Time of day in the hour and minute.
```

Save the ROP or use the following table to record which options are used. For those options that have specific values record those also. The ODD backup schedule will have to be re-entered at the end of the procedure.

Option	Used or unused	Value
FULL		
AM		
CMP		
NRODD		
TO		
RODD		
EVERY	Used	
AT	Used	

OR

```
Response:
    NG - NO SCHEDULE REQUEST
indicating that there is no backup scheduled. Continue at Step 2.4.9.
```

2.4.8 Clear the scheduled automatic ODD Backup. At MCC, type and enter:

```
CLR:ODDBKUP;
```

```
Response: CLR ODDBU COMPLETED
```

2.4.9 Backup the current message class settings. At MCC type and enter:

```
CHG:LPS,MSGCLS=ALL,TOBKUP;
```

```
Response: OK
```

2.4.10 Allow additional fault recovery messages to be printed. At MCC type and enter:

```
CHG:LPS,MSGCLS=ALL,PRINT=ON,LOG=ON;
```

```
Response: OK
```

2.4.11 Inhibit audit for AM (PCD). At MCC type and enter:

```
INH:AUD=PCD,ENV=SMKP;
```

```
Response: OK
```

2.4.12 Inhibit audit for AM (TMSLNK). At MCC type and enter:

```
INH:AUD=TMSLNK,ENV=SMKP;
```

```
Response: OK
```

2.4.13 Inhibit audit for AM (MMPSPYCFG). At MCC type and enter:

```
INH:AUD=MMPSPYCFG,ENV=SMKP;
```

```
Response: OK
```

2.4.14 Inhibit audit for AM (NLI). At MCC type and enter:

```
INH:AUD=NLI,ENV=SMKP;
```

```
Response: OK
```

2.4.15 Inhibit audit for AM (QPIPE). At MCC type and enter:

```
INH:AUD=QPIPE,ENV=SMKP;
```

```
Response: OK
```

2.4.16 Inhibit audit for AM (QLPS). At MCC type and enter:

```
INH:AUD=QLPS,ENV=SMKP;
Response: OK
```

2.4.17 Inhibit audit for AM (QLNK). At MCC type and enter:

```
INH:AUD=QLNK,ENV=SMKP;
Response: OK
```

2.5 Insert Growth SM-XC Data as Unequipped

MGRCV Tool Reference Information

MGRCV File Name	Function	Recent Change View
18_1.grc	Insert	18.1

NOTE: A warning message concerning Facilities Interface Unit (FIU) data being ignored when the SM type is Local Switching Module (LSM) or Host Switching Module (HSM) may be displayed when view 18.1 is inserted. If this warning is displayed, it should be ignored and growth of the SM-XC continued.

2.5.1 Access Recent Change View 18.1 in the Insert mode. Reference Table A.

2.5.2 Select the completed GRCV form 18.1 for the identified SM. From this form, enter the KEY attribute listed in the following display.

RCV 18.1 KEY ATTRIBUTE

*1. SM	___

2.5.3 Using the GRCV 18.1 form as a guide, type and enter the indicated values for each attribute, except set the following attributes to the following specified values:

GRCV Form 18.1

Attribute	Value
UNIT TYPE	SMU6
EQSTAT	U
IIA NORMAL	INHIBIT
VERPOS	See Note
POST PUMP	N
MCTSI 0 CLI	1st unit = 1928; 2nd and 3rd unit = 11928
MCTSI 1 CLI	1st unit = 1928; 2nd and 3rd unit = 11928
DCTURTI	(blank)
SLIM RI	(blank)
ALITRTI	(blank)
PPMTC RI	(blank)
MAJRTI	(blank)
FCLASS	NO
TSISCOM 0 CLI	29
TSISCOM 1 CLI	29
TSIS CLI	20

NOTE: VERPOS is the number of inches the bottom of the unit is off the ground. 1st unit = 6, 2nd unit = 32, 3rd unit = 49

2.5.4 Insert the form.

```
Response: inserting.....FORM INSERTED
```

2.7 Safe Stop Point

2.28 Install the CM 3 Side 0 NLI paddle boards

NOTE: Install the MMA2 or MMA3 NLI paddle boards at the appropriate location on the CM3 indicated in the Office Specification 384. (See Figure 6 for MMA3, or Figure 7 for MMA2)

2.31 Install Loopback Plugs on NCT Links from Side 0 TMS

NOTE: NCT2 Links that terminate to a BKXL1 paddle board will be looped with ITE-7150 Loopback Plugs. NCT2 Links that terminate to a BKXL2 paddle board will be looped with ITE-7093 Loopback Plugs.

1. If NCT Links are not currently disconnected: Disconnect TMS NCT Links at the Side 0 NLI paddle boards. This should be performed twice, once for the even (0A) NLI 00-0 NCT2 Links, and again for the odd (0B) NLI 01-0 NCT Links. See Figure 6 for MMA3/BKXL1, or Figure 7 for MMA2/BKXL2.
2. Install Loopback Plugs. Connect NCT Links (Side 0A and 0B) to the Loopback Plugs.
3. Loop-around continuity should now be established on Side 0 TMS NCT Links.

2.32 Perform NCT Link Loop Test (Side 0 TMS)

This step tests the optical paths between the Side 0 TMS and the growth SM-XC. At MCC, type and enter:

```
DGN:TMSFP=0-0,TMSLNK=a[&&z],PH=7,GROW,RAW,TLP;

Where: a[&&z] = TMS Link number range of growth SM-XC. See Step 2.4.1 for this
          pre-recorded Link information.

Response: DGN TMSFP=0-0 COMPLETED ATP
```

NOTE: An output message will be generated on the ROP which identifies the associated CM3 TMS shelf and circuit pack locations which were tested. This information can be used to replace associated circuit packs when necessary.

Figures

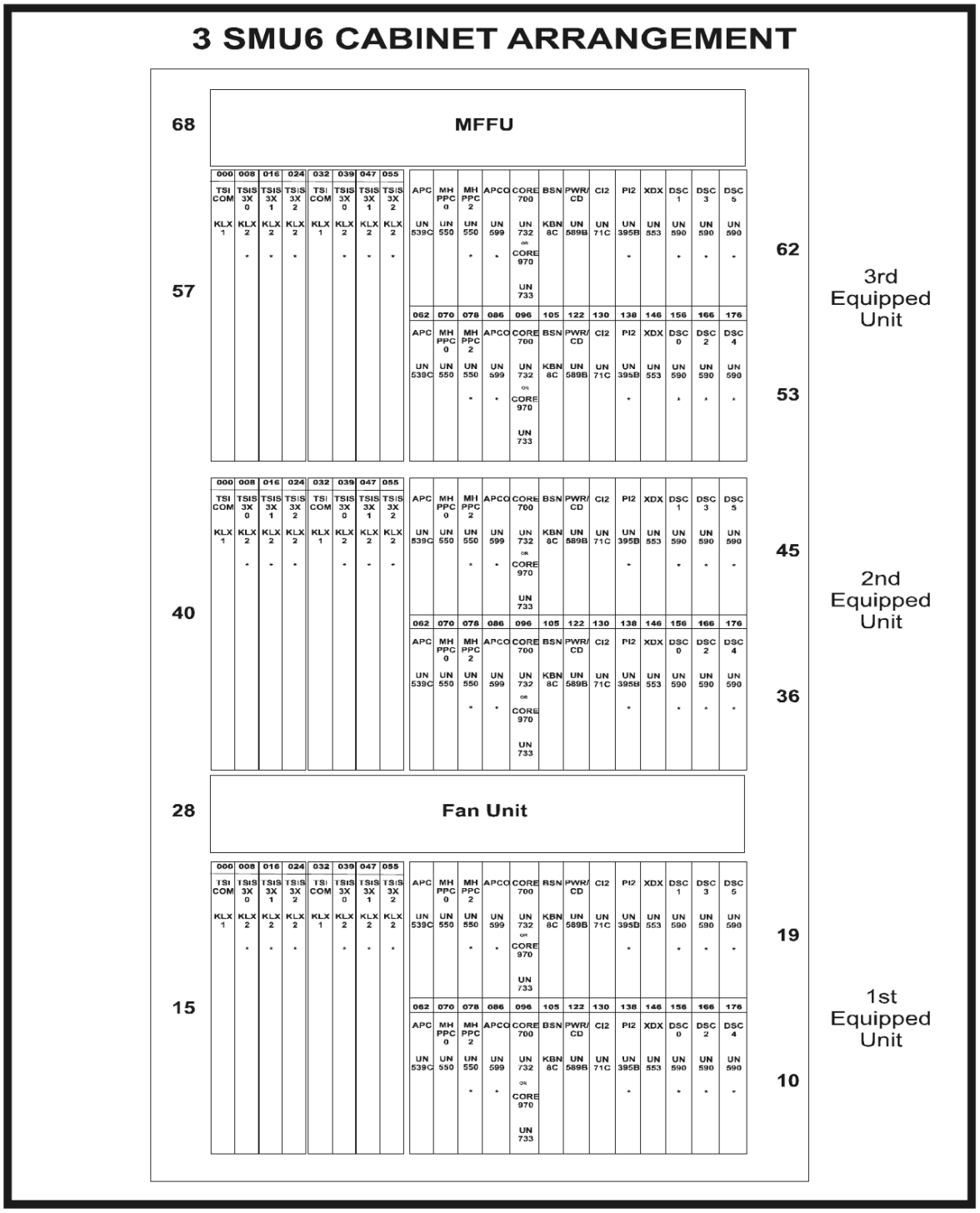


Figure 1. SM-XC Cabinet, 3 SMU6 Configuration

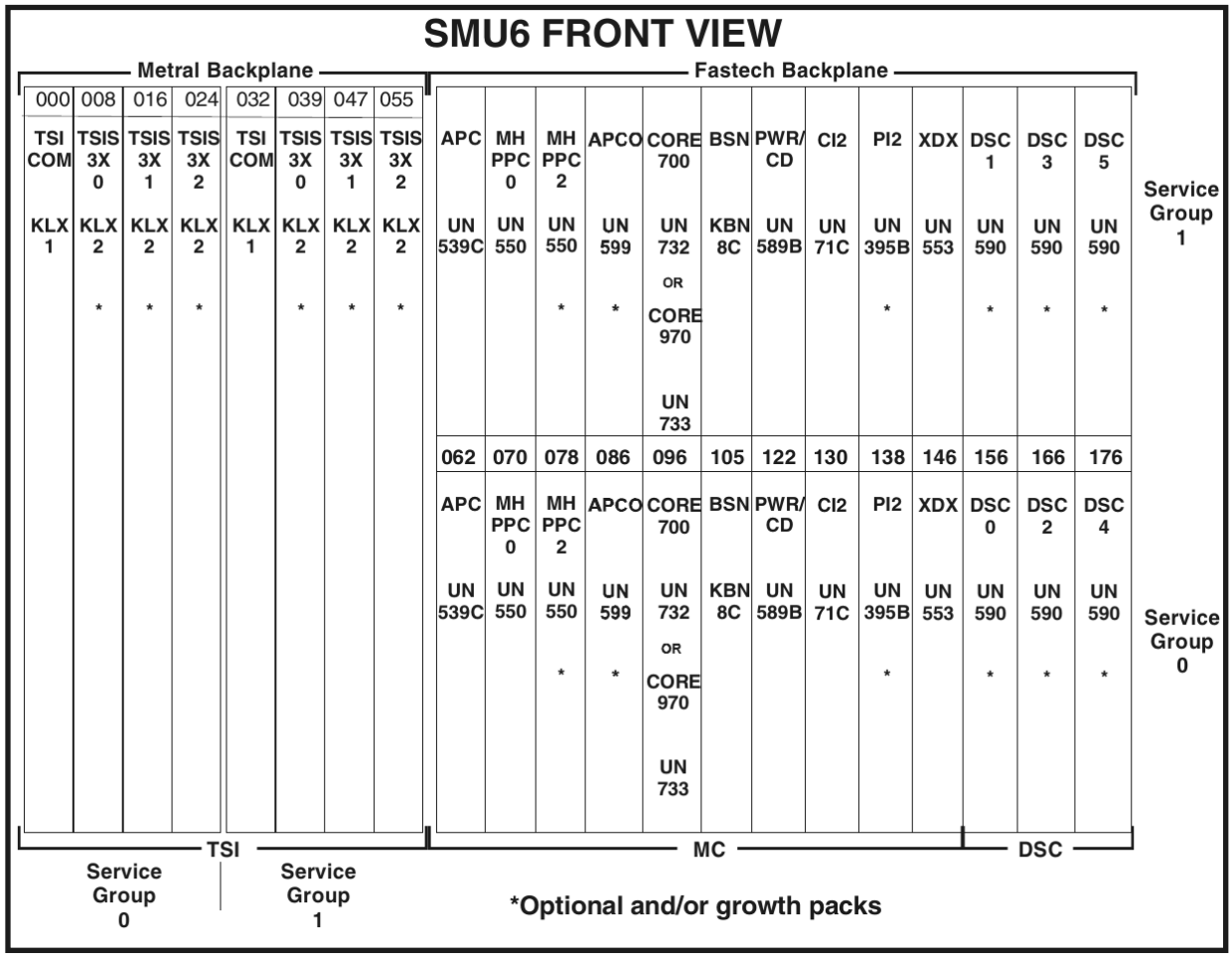


Figure 2. SMU6 Front View

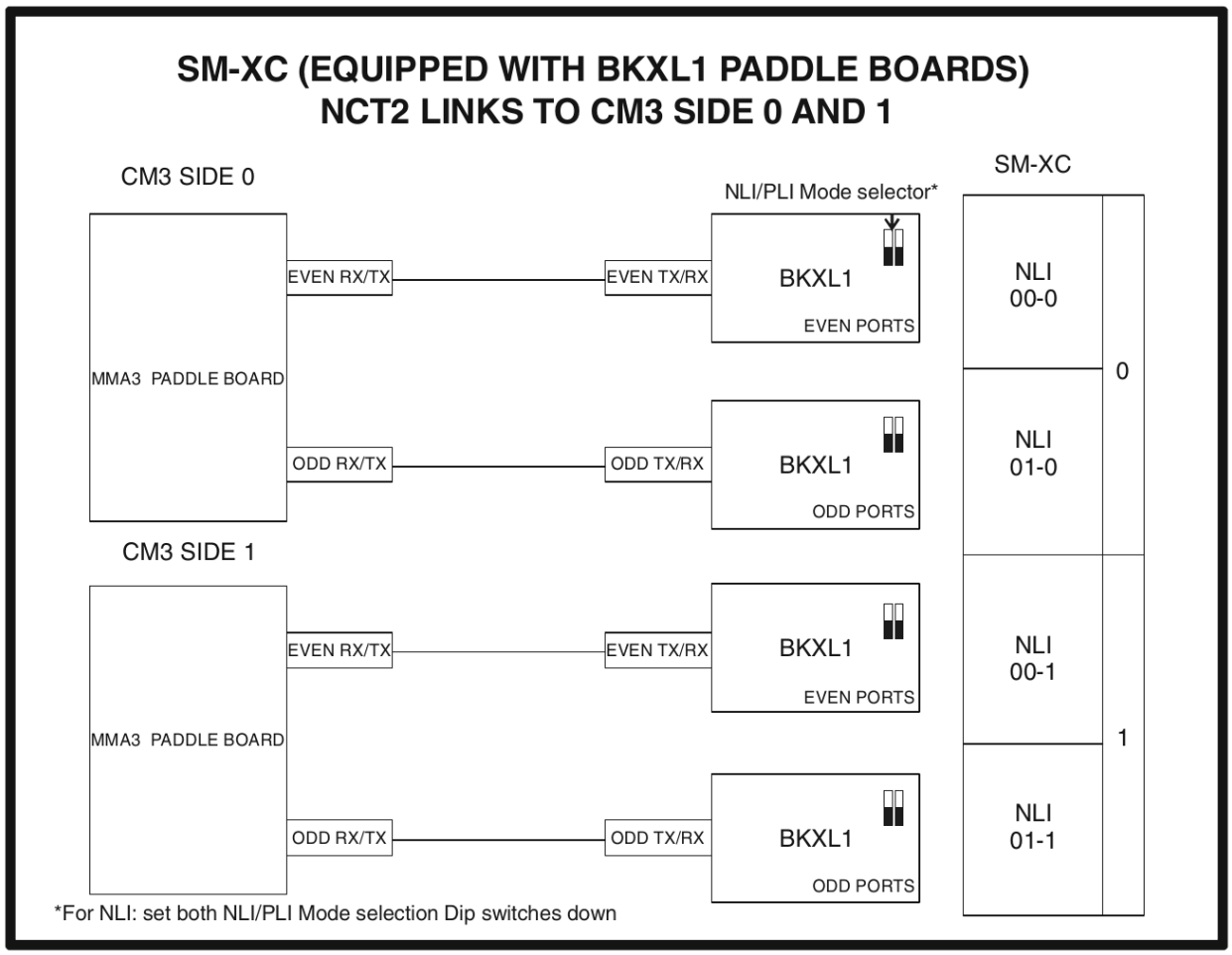


Figure 6. BKXL1 NLIs to CM3 Side 0 and Side 1

Table

Table D. Defined Communication Links(s) Tests

SM-XC CLNKs TESTs - QUAD MMPs ARE ASSIGNED						
ACTION	ACTIVE CLNK STATUS CAN BE DISPLAYED ON MCC page **1900,x** and **126x**					
	PATH 0			PATH 1		
	ONTC	MMP	MSGS	ONTC	MMP	MSGS
Expected CLNK Starting Point	0	0	0	1	1	1
Remove CLNK 111	0	0	0	0	0	0
Restore CLNK 010	0	0	0	0	1	0
Remove CLNK 000	0	1	0	0	1	0
Restore CLNK 001	0	1	0	0	0	1
Remove CLNK 010	0	0	1	0	0	1
Restore CLNK 011	0	0	1	0	1	1
Remove CLNK 001	0	1	1	0	1	1
Restore CLNK 100	1	0	0	0	1	1
Remove CLNK 011	1	0	0	1	0	0
Restore CLNK 110	1	0	0	1	1	0
Remove CLNK 100	1	1	0	1	1	0
Restore CLNK 101	1	0	1	1	0	1
Remove CLNK 110	1	0	1	1	0	1
Restore CLNK 111	1	0	1	1	1	1
Remove CLNK 101	1	1	1	1	1	1
Restore CLNK 000	0	0	0	1	1	1
Restore remaining OOS CLNKs	0	0	0	1	1	1